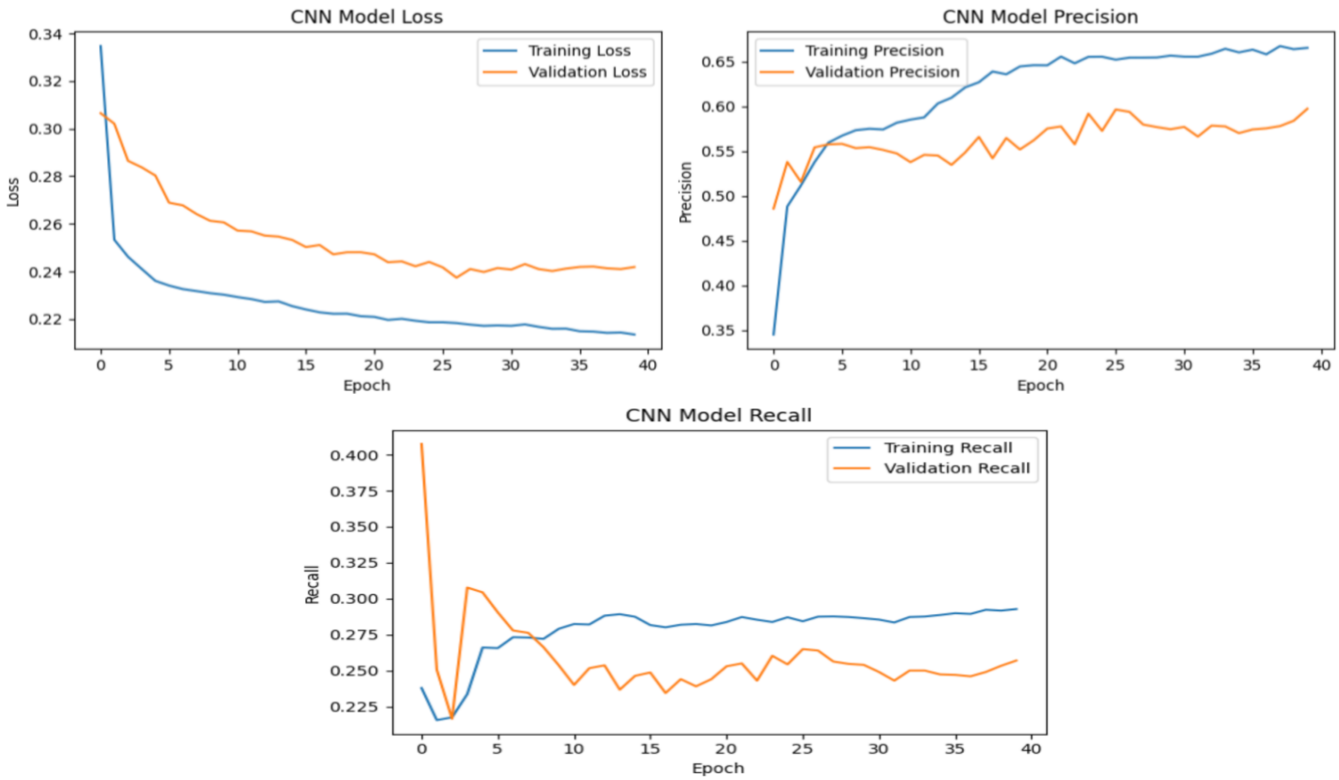
Multimodal Film Genre Classification: CNN and LSTM Model Evaluation

1. Introduction

This report presents an analysis of two deep learning models for multimodal film genre classification. The first model, a Convolutional Neural Network (CNN), identifies film genres based on movie poster images, while the second model uses a Long Short-Term Memory (LSTM) network to classify genres from text film overviews. Both models tackle the same problem through different data modalities. The dataset consists of film posters, text overviews, and genre labels across 25 categories, with each film potentially belonging to multiple genres, making this a multi-label classification task. The models were trained and evaluated using binary cross-entropy loss, focusing on precision and recall as performance metrics. This report analyses the performance of both models, comparing their strengths and weaknesses through examples and evaluating why certain genres are predicted more accurately by one modality over the other.

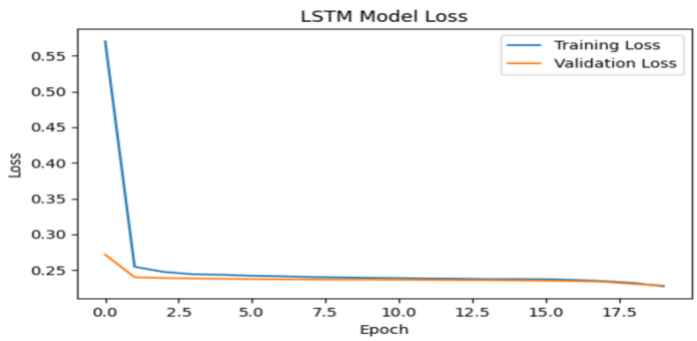
2. CNN Evaluation

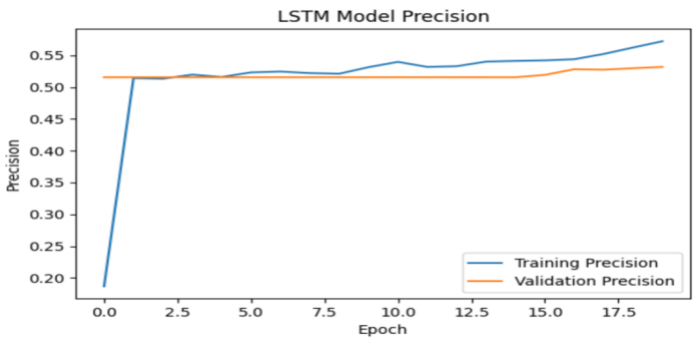
The CNN architecture for poster classification features several **convolutional layers**, **dropout layers**, **max pooling operations**, and **dense layers**. It takes **64x64x3** RGB images (film posters) as input and predicts across **25 genre** categories, with about **3.46 million parameters** to capture visual patterns. Training over **40 epochs** shows **consistent improvement**, with **training loss decreasing** to 0.2131 and **final validation loss** at approximately 0.2419, indicating **moderate overfitting** and **reasonable generalization** to unseen data. **Precision** metrics are **strong**, with training precision at 0.6638 and validation precision around 0.5972, suggesting **reliable predictions** with **fewer false positives**. In contrast, **recall metrics** are **low**, at about 0.2938 for training and 0.2569 for validation, indicating the **model often fails to assign multiple genres** to films. The **precision-recall trade-off** hints that the model is **conservative** in its predictions, **favouring fewer genres** with higher confidence. **The gap** between training and validation metrics is **not severe**, suggesting **good generalizability without severe overfitting.**

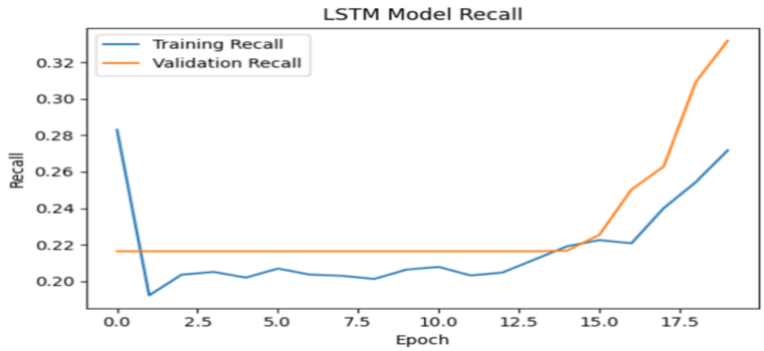


3. LSTM Evaluation

The LSTM model for text-based genre classification has **3.16 million parameters** (12.05 MB) and includes a **TextVectorization layer**, a **256-dimensional embedding layer**, **two bidirectional LSTM layers with dropout**, and **dense layers** for classifying 25 genres. Training metrics show a **significant drop in training loss** from to 0.2606 in the first two epochs, **stabilizing** around 0.23 by epoch 20 suggesting **effective learning with potential saturation**. **Validation loss decreased** from 0.2716 to roughly 0.24 at second epoch and remained **stable** recommending **good generalisation without overfitting**. **Training precision improves** from 0.1375 to approximately 0.5655 by epoch 20 showing **strong classification accuracy gains**, while **validation precision stayed consistent** at around 0.52 to 0.53 pointing to **potential model bias** or **validation data limitations**. Strikingly, **validation recall improved** from about 0.2164 to 0.3319, **surpassing training recall**. This suggests the model is **better generalizing genre identification as training progresses**. The ongoing **gap between precision and recall** indicates that the **LSTM prioritizes precision**, resulting in **fewer false positive assignments** but potentially **missing relevant genres**, a common issue in multi-label classification tasks.







4. Comparison of CNN and LSTM Using Examples

4.1. Film ID: tt0479500

For " **tt0479500**," the film **overview** describes a teenage detective who accompanies her father to Los Angeles, where she investigates a murder mystery involving a movie star. Both **CNN and LSTM models correctly** identified **Comedy** as the primary genre with high confidence scores (CNN: 0.6140, LSTM: 0.5969). This aligns well with the **mystery-comedy nature of the film described in the overview**. However, neither model captured the "**Crime**" aspect mentioned in the ground truth genres, despite the presence of a **murder mystery plot element**.

4.2. Film ID: tt0108320

"**tt0108320**" follows high school graduates in Los Angeles during 1965 as they experience America's changing culture and politics. **Both models accurately classified** this film as **Comedy** and **Drama**, with the **LSTM showing slightly higher confidence** in both genres. Their success in this case demonstrates their ability to **recognize films with mixed tones**, as the **overview suggests** it is a coming-of-age story set against significant historical changes that naturally blend comedic and dramatic elements.

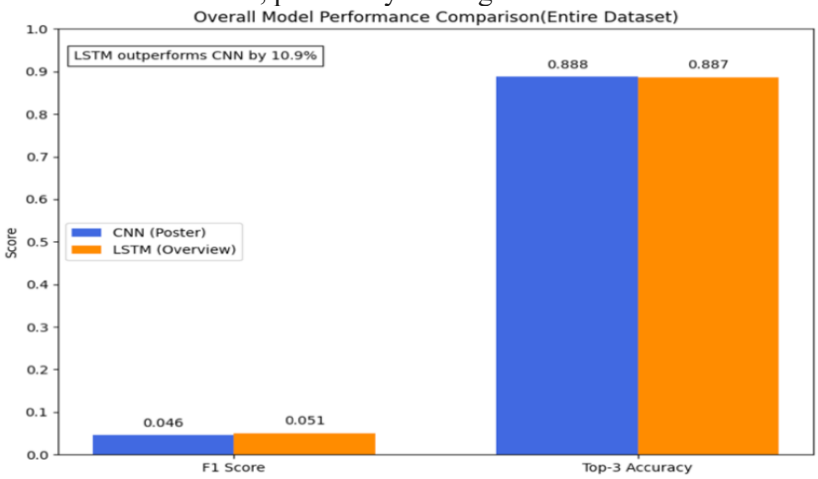
4.3. Film ID: tt0297884

For " **tt0297884**," set in 1950s Connecticut, the **overview** describes a housewife facing a marital crisis amid mounting racial tensions. The **LSTM model outperformed** CNN in genre detection for this film, **correctly identifying both Drama** (0.6125) and **Romance** (0.2961) genres, while **CNN failed** to recognize the **Romance** element, even though the **marital relationship is central** to the plot. This suggests that the **LSTM model** may be **better at capturing the emotional complexity** and **relationship dynamics** described in the overview.

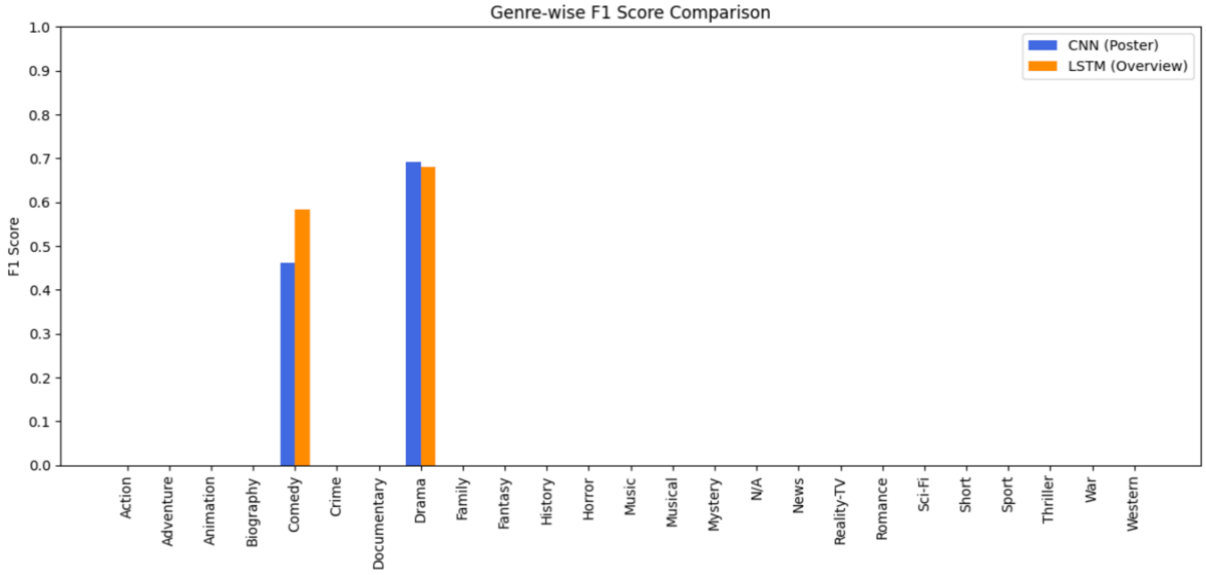
5. Critical Evaluation

Both models have high Top-3 accuracy scores, demonstrating their **ability to identify at least one correct genre in their top three predictions**. The **LSTM outperforms the CNN** by 10.9% in F1 scores (0.051 vs. 0.046), though both scores are relatively low. This gap is not merely technical but reflects more profound insights regarding genre classification and communication. Genres are **fundamentally narrative** and **thematic** constructs, which is why **techniques based on text yield better results than those relying on images**. Movie **posters** are primarily designed for **marketing purposes**, often emphasizing star appeal or visual aesthetics instead of effectively representing the genre. In contrast, **textual summaries clearly outline** the plot and themes, making the genre more explicit. Genre classification

(comedy, romance, drama, action) of **CNN** is influenced by the **dominant colour schemes**, with specific colours representing different genres (for example, **vibrant hues for comedy** and **warm tones for romance**). However, relying heavily on colour **can introduce bias**, especially when genres overlap in their colour palettes. Moreover, **information loss** during pooling or compression can obscure subtle visual details important for distinguishing between genres, particularly in low-contrast scenes like dramas, potentially leading to decreased model accuracy.



The **genre-wise F1 score comparison** shows that **both models perform well** in classifying **Drama**, with CNN achieving about 0.69 and LSTM slightly lower at 0.67. However, in **Comedy**, the LSTM (0.58) significantly outperforms the CNN (0.46). It is also notable that many other genres are not represented, indicating **both models struggle outside of Comedy and Drama**. The models were trained using a **particular set of film data**, which may **restrict their ability** to adjust to **evolving genre standards** and the blending of styles in **contemporary cinema**. Additionally, the **dataset is imbalanced**, with some genres receiving less representation, which could affect the results of the models.



Analysing sample films, **strong Drama performance of models** aligns with accurate predictions for "tt0297884" and "tt0108320." The **strength of LSTM in Comedy** is evident with "tt0479500" and "tt0108320." However, the **lack of F1 scores** for **Romance** highlights their struggle with romantic elements in " tt0297884." The **contrast between low F1 scores** and **high Top-3 accuracy** suggests the models can **recognize relevant genres** yet **often rank them inaccurately** or **lack confidence**. While they **identify genre features in posters and text**, defining **precise genre** boundaries **remains a challenge**, especially for **less common categories**. **The CNN model could improve** by employing transfer learning with pre-trained models from ImageNet to better capture basic visual features. **Enhancements to the LSTM model** might include integrating attention mechanisms to emphasize genre-related phrases or utilizing transformer architectures known for effective text classification. Interestingly, **neither model currently combines visual and textual data**; adopting a multimodal approach that integrates both could significantly boost classification accuracy.

6. Conclusion

This analysis shows that LSTM models significantly outperform CNN models in classifying movie genres based on text from movie overviews. While movie posters contain genre-related visual cues, they can be ambiguous, whereas overviews provide clear, relevant information. There are instances, particularly with visually distinct genres where visual cues may be more reliable. Therefore, a comprehensive classification system should integrate both text and visual approaches.